第二十八讲

作业：14-7(b),9(a),14,11,
15-15

一、平面刚架空间受力

二、思考题讨论

利用基本系统 → 静不定结构的位移

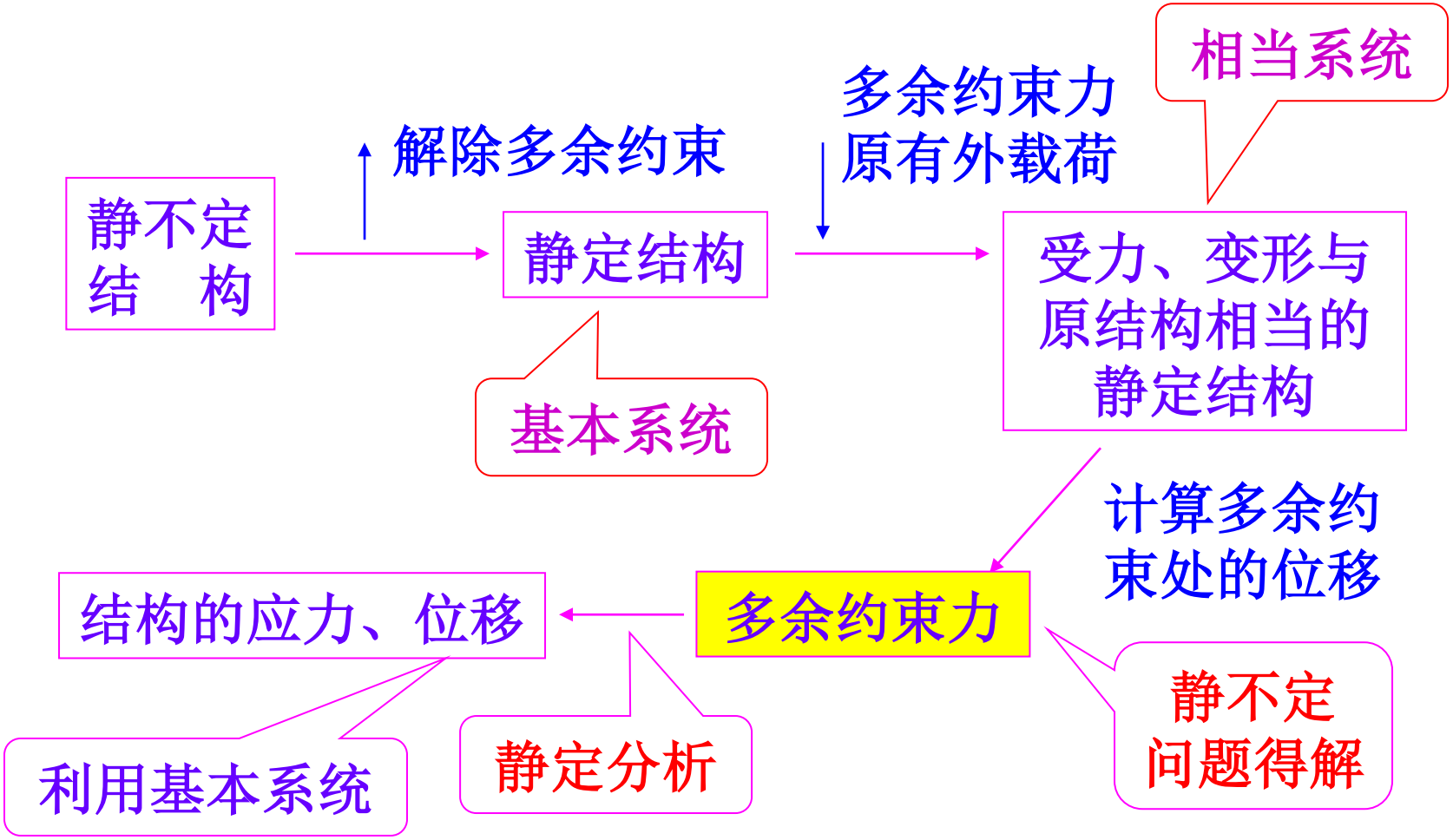
三、能量法与静不定例题

第二十八讲（3）

能量法与静不定例题

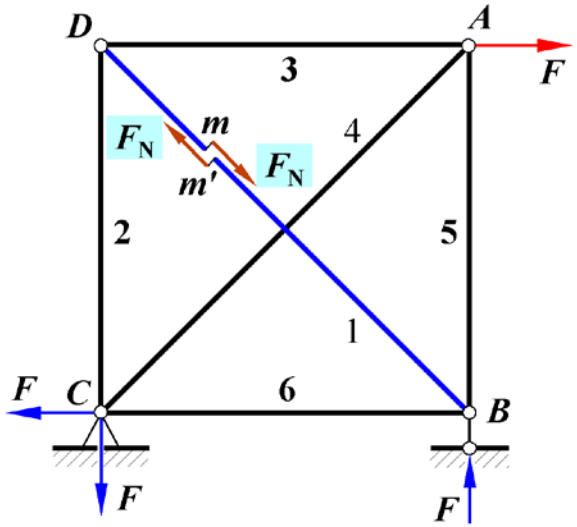
作业：14-7(b),9(a),14,11,
15-15

回顾：力法求解思路


静不定结构
解除多余约束
静定结构
基本系统
多余约束力
原有外载荷
相当系统
受力、变形与原结构相当的静定结构
计算多余约束处的位移
多余约束力
静不定问题得解
静定分析
结构的应力、位移
利用基本系统

思考

可否去掉DB杆建立相当系统？
相应的变形协调条件是什么？

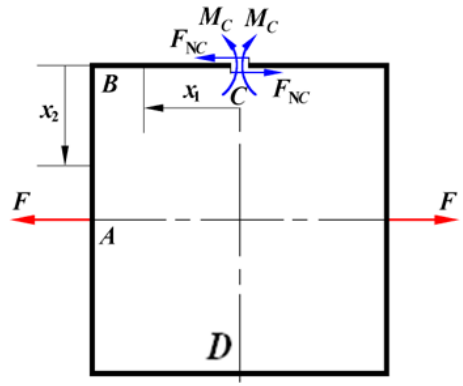


$$\Delta_{D/B} = -\Delta l_1$$

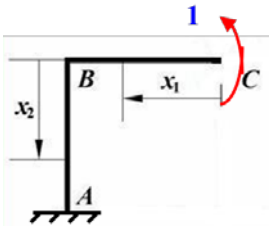
$$\Delta_{D/B} = \sum_{i=2}^{6} \frac{\bar{F}_{Ni1} F_{Ni} l_i}{E_i A_i} = -\frac{F_N l_1}{EA}$$

对称面视为固定端

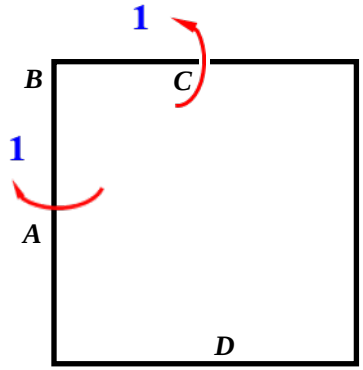
是否可用？

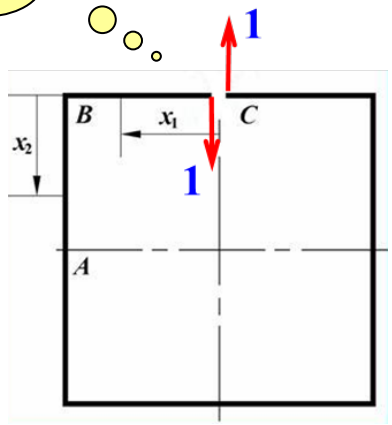


相当系统



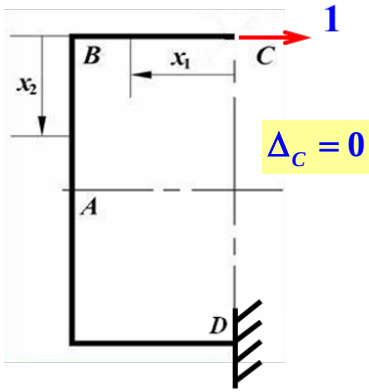
$\theta_C = 0$



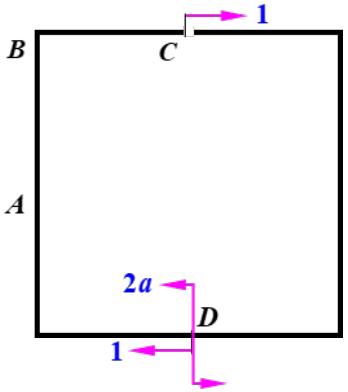


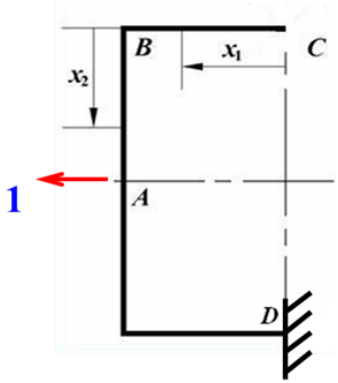
$f_{C^-/C^+} = 0$

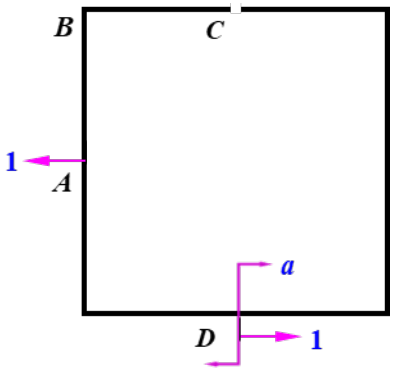
$$1\times\theta_C + 1\times\theta_A = \int_l \frac{\overline{M}(x)M(x)}{EI}\mathrm{d}x$$



$\Delta_C = 0$







$\theta_D = 0$

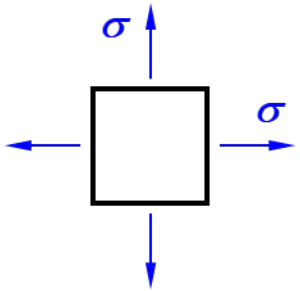
$$1\times\Delta_{C/D} + 2a\times\theta_D = \int_l \frac{\overline{M}(x)M(x)}{EI}\mathrm{d}x$$

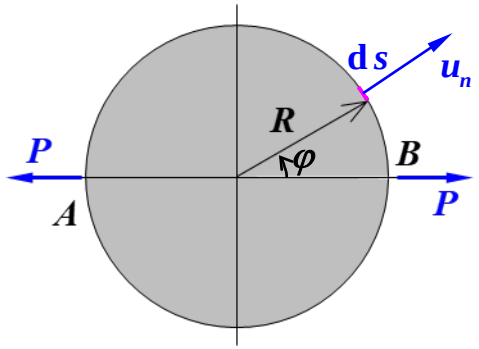
$$1\times\Delta_{A/D} + a\times\theta_D = \int_l \frac{\overline{M}(x)M(x)}{EI}\mathrm{d}x$$

功的互等定理

例 图示匀质等厚圆板，半径R，厚度b，弹性模量E，泊松比 μ，沿直径受一对力P作用。求面积变化ΔA.

解：
$$\Delta A = \oint u_n \, \mathrm{d}s$$

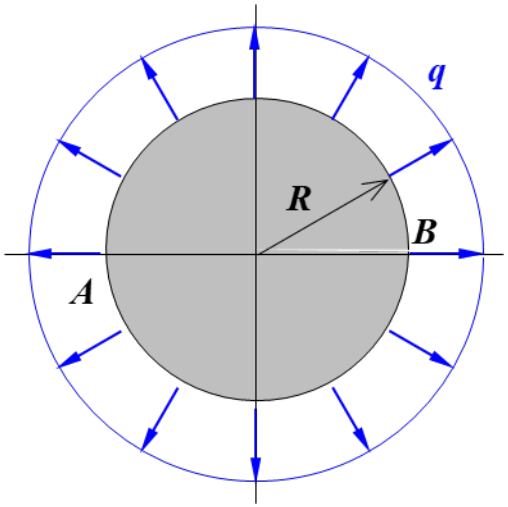




构造均布法向载荷

$$\boxed{P\Delta D_{AB} = \oint u_n q \, \mathrm{d}s}$$

轴对称受力，二向等拉



$$\sigma = \frac{q}{b}, \quad \varepsilon = \frac{1-\mu}{E}\sigma \qquad \Delta D_{AB} = 2R\varepsilon$$

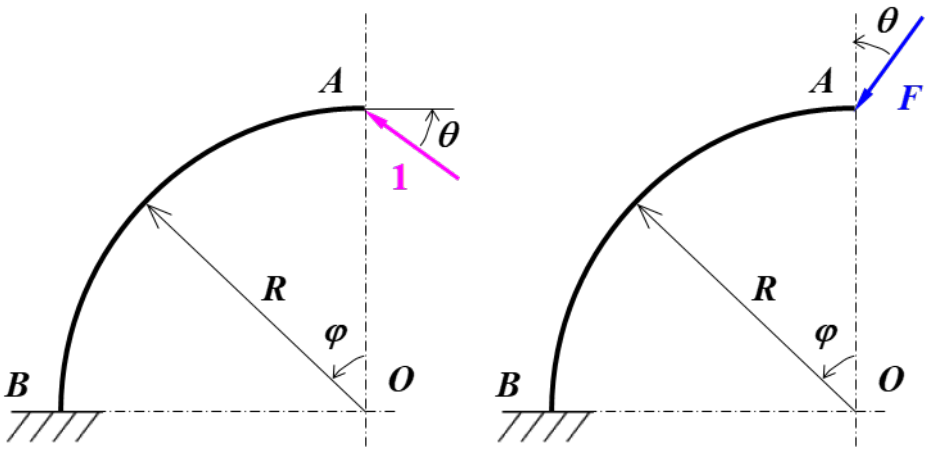
$$\Delta A = \frac{2(1-\mu)PR}{Eb}$$

单位载荷法应用

例 图示1/4圆弧小曲率杆*AB*，*EI*为常数，半径为*R*，自由端受集中力*F*作用。为使*A*截面的总位移沿*F*方向，θ=?

$$M(\varphi)=F\cos\theta R\sin\varphi$$
$$-F\sin\theta R(1-\cos\varphi)$$

$$\overline{M}(\varphi)=-\sin\theta R\sin\varphi$$
$$-\cos\theta R(1-\cos\varphi)$$

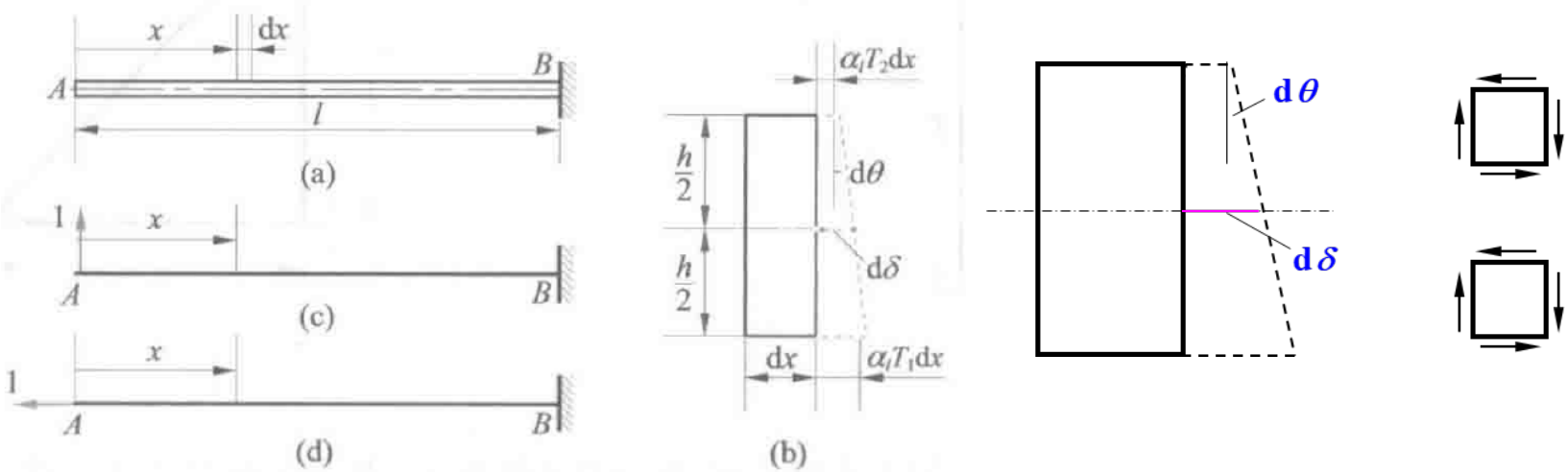


$$\Delta_{AF}=\int_l\frac{\overline{M}(\varphi)M(\varphi)}{EI}R\,\mathrm{d}\varphi=0$$

$$\tan 2\theta=\frac{2}{\pi-4}\qquad\theta=-66.7°$$

梁为什么会弯

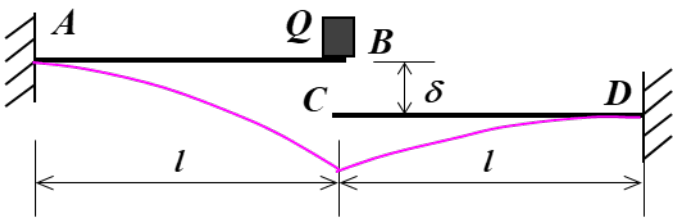
习题13-31 温度导致纵向变形不一致



- 温度变化 → 纵向线性伸长 → 截面转动 → 切应变；
- 维持该切应变的切应力不平衡 → 无切应变；
- 轴线变弯 → 截面与轴线正交。
- 若温度变化上下对称，是否变弯？

分段冲击问题

例 图示悬臂梁AB与CD相同，EI为常数，两自由端B与C间距为 $\delta = Ql^3/3EI$，重物Q突然施加于B处，求：C截面的挠度f_C.



对于AB梁

$$K_d = 1 + \sqrt{1 + \frac{2h}{\Delta_{st}}} = 2$$

$$\Delta_d = 2\frac{Ql^3}{3EI} > \delta$$

二梁接触

$$Q(\delta + f_C) = \frac{k_{AB}}{2}(\delta + f_C)^2 + \frac{k_{CD}}{2} f_C^{\,2}$$

- 绕过冲击载荷
- 以位移表示变形能
- 能量守恒

$$k_{AB} = k_{CD} = \frac{3EI}{l^3}$$

$$f_C = \frac{\sqrt{2}}{2}\delta$$

$$V_\varepsilon = \frac{k\Delta^2}{2} = \int_l \frac{M^2(x)}{2EI}\mathrm{d}x = \frac{F_d\Delta_d}{2} = \frac{F_d^2}{2k}$$

本讲：平面刚架空间受力

下讲：疲劳、截面剪心